where the asymptotic error analysis (for $h \to 0$) may be less relevant. Nevertheless there are applications in molecular modelling where higher accuracy is needed in certain parts of a calculation in order to provide optimal sampling. An example will be found in the isokinetic stochastic integrator of Chap. 8.

For more discussion of higher order integrators, refer to [139, 164, 227, 259, 260, 262, 277, 290]. Of particular relevance for molecular dynamics is the excellent numerical study of [153].

3.3.4 The Takahashi-Imada Method

Recall the Takahashi-Imada method introduced in the last chapter,

$$\begin{aligned}\hat{\boldsymbol{P}} &:= \boldsymbol{p} - (h/2)\nabla\tilde{U}(\boldsymbol{q}),\\ \boldsymbol{Q} &:= \boldsymbol{q} + h\boldsymbol{M}^{-1}\hat{\boldsymbol{P}},\\ \boldsymbol{P} &:= \hat{\boldsymbol{P}} - (h/2)\nabla\tilde{U}(\boldsymbol{Q}),\end{aligned}$$

with modified potential energy function

$$\tilde{U}(\boldsymbol{q}) = U(\boldsymbol{q}) - \frac{h^2}{24}\nabla U(\boldsymbol{q})^T\boldsymbol{M}^{-1}\nabla U(\boldsymbol{q}).$$

One might recognize the modification as being proportional to one part of the commutator expansion in the Verlet method, in fact

$$\nabla U(\boldsymbol{q})^T\boldsymbol{M}^{-1}\nabla U(\boldsymbol{q}) = \{U,\{U,T\}\}.$$

This is certainly not a coincidence. If we replace the potential U by $\tilde{U}$ in the Verlet expansion, we have

$$\begin{aligned}\tilde{H}_h &= T + \tilde{U} + \frac{h^2}{12}\left(\{T,\{T,\tilde{U}\}\} - \frac{1}{2}\{\tilde{U},\{\tilde{U},T\}\}\right) + \mathcal{O}(h^4)\\ &= T + U - \frac{h^2}{24}\{U,\{U,T\}\} + \frac{h^2}{12}\left(\{T,\{T,\tilde{U}\}\} - \frac{1}{2}\{\tilde{U},\{\tilde{U},T\}\}\right) + \mathcal{O}(h^4)\\ &= H + \frac{h^2}{12}\left(\{T,\{T,\tilde{U}\}\} - \{\tilde{U},\{\tilde{U},T\}\}\right) + \mathcal{O}(h^4)\\ &= H + \frac{h^2}{12}\left(\boldsymbol{p}^T\boldsymbol{M}^{-1}U''\boldsymbol{M}^{-1}\boldsymbol{p} - \nabla U^T\boldsymbol{M}^{-1}\nabla U\right) + \mathcal{O}(h^4).\end{aligned}$$

Introducing coordinate transformations